

Improved Quality Diversity for Noisy Fitness Functions using Hypothesis Testing Replacement

Abstract

Quality-Diversity (QD) algorithms excel at discovering a repertoire of both high-quality and behaviourally diverse solutions. However, when fitness functions are stochastic, such as multi-rater or generative model-based feedback, traditional MAP-Elites frameworks suffer from severe score inflation and archive drift. We define score inflation as a positive selection bias where lower-quality candidates are erroneously accepted into the archive by exploiting high-variance, lucky noise spikes in their evaluations. To eliminate these stochastic artifacts, we introduce HT-MAP-Elites. This framework replaces raw mean-score comparisons with a strict, hypothesis-testing replacement gate, separating exploratory candidate sampling from statistical archive validation.

We evaluate HT-MAP-Elites against a standard MAP-Elites baseline across 15 independent seeds. To prove score inflation, we introduce a high-sample ground-truth re-evaluation harness ($M_{\text{test}} \geq 100$) on the finalized archives. Through a controlled noise sweep, we show that naive MAP-Elites suffers from systematic score inflation that grows with evaluation noise - its archive quality increasingly overstates its true quality as rater variance rises. The hypothesis-testing gate significantly reduces this inflation at every noise level tested ($p \leq 0.001$), yielding archives whose reported quality more faithfully reflects true quality. We further show that single-evaluation gating can be overly conservative, and that this exploration-validation trade-off is governed by the evaluation budget allocated at the gate.

Rather than claiming uniform dominance, our results characterise when and how hypothesis-testing replacement is beneficial under stochastic feedback.

1. Introduction

Recent advances in generative models have allowed high quality image synthesis through flexible, black-box objectives. To optimize these models towards target distributions, researchers increasingly leverage external evaluators, such as VLMs as proxies for human reference.

A promising direction for generative discovery is Quality-Diversity (QD) search, which aims to discover a repertoire of both high-performing and behaviourally diverse solutions within a structured feature space. Algorithms such as MAP-Elites have been successfully deployed to illuminate behavioural niches by maintaining an archive of "elite" solutions across a grid of behavioural descriptors. However, standard QD formulations operate under a critical assumption that the black-box evaluation metric provides a reliable, deterministic evaluation of a candidate's underlying quality.

In practice, this assumption would not hold fully when using generative models to provide fitness scores, since their outputs are highly stochastic and often come from multiple raters. As a result, the true quality of a candidate is not a single scalar, but a distribution of scores. Despite this fact, existing QD methods rely on naive archive replacement rules, typically comparing only the mean score of candidates. If the distribution is sufficiently wide, the

incumbents may be replaced with lower-performing solutions, and the final archive may consequently be sub-optimal.

In this work, we address this vulnerability by introducing HT-MAP-Elites, a distributionally robust archive update mechanism engineered for stochastic generative optimization. Crucially, our approach does not alter the underlying generative processes or modify the rater architectures; instead, it targets the decision layer of QD containers, ensuring seamless compatibility with existing hardware-accelerated evolutionary pipelines. We evaluate HT-MAP-Elites against a traditional MAP-Elites baseline across 15 independent seeds. We demonstrate that our hypothesis testing framework removes score inflation and preserves a significantly more stable and high quality repertoire of solutions.

2. Related Works

The development of this work sits at the intersection of quality-diversity search, query-efficient diffusion guidance and statistical optimization.

2.1. Quality-Diversity and MAP-Elites

Quality-Diversity algorithms aim to maintain a collection of high performing and diverse set of solutions, rather than a single global optimum. The MAP-elites algorithm (Mouret and Clune, 2015) serves as the foundation framework for our approach, transforming a behavioural space into a grid of niches where only the “elite” (i.e. the highest performing solution) of each cell is preserved. While this approach is traditionally applied to robotics and neuroevolution, we extend this logic to generative diffusion models.

2.2. QDax: Accelerated Quality-Diversity

The core implementation is built upon QDax (Lim et al, 2022), a framework that leverages JAX for hardware-accelerated QD search.

2.3. Uncertain Quality-Diversity

Flageat and Cully (2023) formalise the Uncertain QD setting, in which the fitness and behavioural descriptor of each solution are no longer fixed scalars but distributions over possible values. Their diagnosis of standard MAP-Elites under stochastic evaluation aligns with ours: elitist replacement rules systematically retain “lucky” solutions whose single observed fitness overestimates their true expected quality, leading to archives populated by sub-performing outliers rather than genuinely high-quality elites. They quantify this through a QD-score-Loss metric, which is a normalised gap between the archive’s reported QD score and a corrected score obtained via evaluation, which is conceptually very similar to inflation gap. Both metrics capture the same underlying phenomenon: noise-driven overestimation of archive quality that worsens as evaluation variance increases.

3. Preliminaries

We first introduce a few notations and key concepts as a background.

3.1. Stochastic Black-Box Evaluation

Traditional optimisation assumes an objective function $f(\mathbf{x})$ that returns a reliable scalar. When using generative models, the feedback is inherently stochastic. We model the rating from a model v on image $\mathbf{x}$ as:

$$r_v(\mathbf{x}) = -\mathbf{w}_v^\top \boldsymbol{\psi}(\mathbf{x}) + \varepsilon_v, \quad \varepsilon_v \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where $\mathbf{w}_v \in \mathbb{R}^5$ is model v 's fixed, ℓ_2 -normalised preference vector over five perceptual image features captured by $\boldsymbol{\psi}(\mathbf{x}) \in [0, 1]^5$, and ε_v models inter-query variance. The negation makes this a minimisation problem: a model that strongly prefers a feature assigns a *lower* (better) score to images that exhibit it.

Per evaluation, M models are sampled uniformly without replacement from a pool of N_V , yielding a rating vector:

$$\mathbf{r}(\mathbf{x}) = [r_{v_1}(\mathbf{x}), \dots, r_{v_M}(\mathbf{x})] \in \mathbb{R}^M, \quad \{v_1, \dots, v_M\} \subset \{1, \dots, N_V\}. \quad (2)$$

Because the same image evaluated multiple times yields different scores through different sampled subsets and independent noise draws, the true quality of a candidate is not a single scalar but a distribution. Existing methods rely on naive mean comparison $\bar{r}_{\text{new}} < \bar{r}_{\text{old}}$ to guide archive updates, conflating objective signal with rater noise. This leads to *archive drift*, where high-quality incumbents are erroneously replaced by noisy challengers.

3.2. MAP-Elites and Behavioural Spaces

Quality-Diversity (QD) algorithms search for a diverse set of high-performing solutions within a structured behaviour space. We define a behaviour descriptor function $\boldsymbol{\phi}(\mathbf{x}) \in [0, 1]^2$ that maps an image to a 2D coordinate based on intrinsic properties (e.g., brightness and entropy). This space is discretized into a grid $\mathcal{C}$ of $N \times N$ cells. In the standard MAP-Elites framework, each cell c stores a single elite $\mathbf{x}_c$.

The scalar fitness stored per cell is defined as the negated mean rating, such that QDax's maximisation convention aligns with score minimisation:

$$f(\mathbf{x}) = -\frac{1}{M} \sum_{m=1}^M r_{v_m}(\mathbf{x}) = -\bar{r}(\mathbf{x}) \quad (3)$$

Higher fitness therefore corresponds to a lower mean model rating, i.e., higher perceived image quality. The overall Quality-Diversity score aggregates fitness across all occupied cells:

$$\text{QD} = \sum_{\substack{c \in \mathcal{C} \\ A[c] \neq \emptyset}} f(x_K^c) \quad (4)$$

A high QD score indicates that many behavioural niches are simultaneously occupied by high-quality elites. The baseline replacement rule is naive: a challenger $\mathbf{x}_{\text{new}}$ replaces the incumbent $\mathbf{x}_c$ if $f(\mathbf{x}_{\text{new}}) > f(\mathbf{x}_c)$, equivalently $\bar{r}_{\text{new}} < \bar{r}_{\text{old}}$. While effective in deterministic environments, this rule is highly vulnerable to the stochastic model noise defined in Section 3.2, and serves as the primary baseline for our ablation studies.

3.3. Statistical Decision Metrics

To stabilize the repertoire against stochastic evaluation noise, HT-MAP-Elites utilises three distinct statistical metrics to govern the archive replacement logic: the Common Language Effect Size (CLES), an exact two-sample Z -test, and Cohen’s d .

Common Language Effect Size (CLES). The CLES $\hat{\delta}$ represents the empirical probability that a randomly drawn rating from the challenger’s distribution is lower (better) than a random draw from the incumbent’s:

$$\hat{\delta} = \frac{1}{M^2} \sum_{i=1}^M \sum_{j=1}^M \left[\mathbb{I}(r_i^{new} < r_j^{old}) + \frac{1}{2} \mathbb{I}(r_i^{new} = r_j^{old}) \right] \quad (5)$$

A value of $\hat{\delta} = 0.5$ indicates performance at parity with chance, while $\hat{\delta} = 1.0$ indicates strict dominance. HT-MAP-Elites enforces a threshold of $\hat{\delta} > \delta_{\min} = 0.55$ to ensure that archive updates represent practically meaningful improvements rather than marginal gains within the noise floor.

Exact Z -test Verification. Unlike naive mean comparison, this test accounts for the rater count M and the variance within score distributions. The test statistic utilizes the standard error of the difference between two sample means:

$$z = \frac{|\bar{r}_{new} - \bar{r}_{old}|}{\sqrt{\frac{\sigma_{new}^2}{M} + \frac{\sigma_{old}^2}{M} + \epsilon}} \quad (6)$$

The corresponding two-tailed p -value is calculated using the Gaussian cumulative distribution function (Φ):

$$p = 2(1 - \Phi(z)) \quad (7)$$

An update to the robust archive is accepted only if $p < \alpha$ (where $\alpha = 0.10$). This approach enables statistically grounded decisions entirely within the JAX-native framework, maintaining exactness relative to the environment noise scale $\sigma = 0.05$.

Cohen’s d . Distinct from CLES, Cohen’s d is used as a post-hoc diversity metric to quantify the magnitude of difference between discovered behavioral niches. For any two occupied cells A and B with rating distributions r_A and r_B :

$$d_{AB} = \frac{\bar{r}_A - \bar{r}_B}{\sqrt{(\sigma_A^2 + \sigma_B^2)/2}} \quad (8)$$

How the metrics are used While the Z -test and CLES governs online replacement decisions, Cohen’s d characterizes the final repertoire’s quality spread. A high mean $|d_{AB}|$ across cell pairs indicates that the HT replacement rule has successfully preserved genuinely distinct solution profiles rather than noisy variants of the same candidate.

4. Proposed Method: HT-MAP-Elites

The HT-MAP-Elites framework stabilizes Quality-Diversity (QD) search under stochastic feedback by utilizing a concurrent, dual-archive *Leader-Follower* architecture. This design decouples exploratory behavioral discovery from high-precision statistical validation, ensuring that the final repertoire is robust to the rater noise defined in Section 4.2.

4.1. Concurrent Leader-Follower Architecture

Unlike traditional MAP-Elites which utilizes a single repertoire, HT-MAP-Elites maintains two distinct archives that operate in parallel throughout the optimization process. This architecture prevents the “Zero-Progress” trap—where a conservative statistical test might otherwise lock out valid new discoveries during the early stages of search.

- **Exploratory Scout Archive (A_1):** The Scout archive prioritizes rapid behavioral discovery. It utilizes a naive mean-based replacement rule, accepting any candidate that provides a directional improvement in mean model rating. This archive serves as a high-turnover sampler that maintains search momentum and provides a diverse pool of candidates for validation.
- **Robust Judge Archive (A_2):** The Judge archive serves as the final, verified high-performance archive. Updates to A_2 are intermittent and governed by a strict hypothesis testing (HT) rule. This archive only accepts candidates that demonstrate statistically significant improvements over incumbents, effectively filtering out stochastic noise artifacts.

4.2. Distributional Refinement and Refine Frequency

The interaction between the two archives is governed by a *Refine Frequency* parameter (f_{refine}). While A_1 is updated every iteration, A_2 only pulls the current best-performing elites from A_1 every f_{refine} iterations. This synchronization delay allows the Scout to explore a wider search horizon before the Judge performs a rigorous assessment, creating a stable “Leader-Follower” lag that preserves high-confidence discoveries.

When the Judge receives a candidate from the Scout, it does not reuse the Scout’s stored scores. Instead, it queries the model jury again to obtain a fresh, independent set of ratings. This is necessary because the Scout’s stored scores are the very scores that caused the candidate to be accepted — they are biased in the candidate’s favour. By re-evaluating with independent noise, the Judge ensures its replacement decision is based on unbiased evidence.

4.3. Hypothesis-Testing (HT) Replacement Rule

When a challenger candidate x_{new} from the Scout archive attempts to displace an incumbent x_{old} in the Judge archive, it must pass a three-condition HT rule. We treat the rater scores $r(x)$ as a distribution and calculate the standard error (SE) of the difference between the sample means:

$$SE = \sqrt{\frac{\sigma_{new}^2}{M} + \frac{\sigma_{old}^2}{M}} \quad (9)$$

The decision is governed by the following criteria:

1. **Directional Improvement:** The challenger’s mean rating must be lower than the incumbent’s: $\bar{r}_{new} < \bar{r}_{old}$

2. **Statistical Significance** ($p < \alpha$): We utilize an exact two-tailed Z -test. The p -value is calculated using the Gaussian cumulative distribution function (Φ):

$$z = \frac{|\bar{r}_{new} - \bar{r}_{old}|}{SE + \epsilon}, \quad p = 2(1 - \Phi(z)) \quad (10)$$

An update is only accepted if p falls below the significance threshold α (typically 0.10).

3. **Practical Significance** ($\hat{\delta} > \delta_{\min}$): To ensure improvements are not just statistically significant but also practically meaningful, we calculate the Common Language Effect Size (CLES). The CLES represents the probability that a random draw from the challenger's scores is superior to one from the incumbent's:

$$\hat{\delta} = \frac{1}{M^2} \sum_{i=1}^M \sum_{j=1}^M \left[\mathbb{1}(r_i^{new} < r_j^{old}) + \frac{1}{2} \mathbb{1}(r_i^{new} = r_j^{old}) \right] \quad (11)$$

The replacement is permitted only if $\hat{\delta}$ exceeds the minimum effect size floor $\delta_{\min}$ (typically 0.55).

Only when all three conditions are met does the Judge archive permit the update, ensuring that the final repertoire remains distributionally robust.

Algorithm 1: HT-MAP-Elites

1. Initialize $\mathcal{A}_1$ (Scout) and $\mathcal{A}_2$ (Judge) as empty distributional archives.
 2. Initialize fixed model weights $W \in \mathbb{R}^{N_V \times 5}$ and JAX PRNG key.
 3. **for** $t \leftarrow 1$ **to** T **do**
 4. // 1. Scout Exploration (Archive $\mathcal{A}_1$)
 5. Sample batch U of B genotypes (mutations from $\mathcal{A}_1$ or fresh samples).
 6. Generate phenotypes $X = S_\theta(U)$ and extract descriptors $\phi(X)$.
 7. Query Model Jury: $R_{new} \in \mathbb{R}^{B \times M}$ where $r_{b,m} = -w_m^\top \psi(x_b) + \epsilon$.
 8. **for** each candidate $b \in B$ **do**
 9. Map to cell $c_b \leftarrow \lfloor \phi(x_b) \times N \rfloor$.
 10. **if** $\bar{r}_b^{new} < \bar{r}_b^{old}$ in $\mathcal{A}_1[c_b]$ **then** Update $\mathcal{A}_1[c_b] \leftarrow (u_b, x_b, r_b^{new})$.
 11. // 2. Judge Refinement (Archive $\mathcal{A}_2$)
 12. **if** $t \pmod{f_{refine}} = 0$ **then**
 13. **for** each occupied cell c in $\mathcal{A}_1$ **do**
 14. Retrieve challenger r_{A1}^c and incumbent r_{A2}^c .
 15. Increment *attempt_counts*[c].
 16. **if** $\mathcal{A}_2[c]$ is empty **then**
 17. $\mathcal{A}_2[c] \leftarrow \mathcal{A}_1[c]$; Increment *acceptance_counts*[c].
 18. **else**
 19. $(\text{accept}, p, \delta) \leftarrow \text{HT}(r_{A1}^c, r_{A2}^c, \alpha, \delta_{\min})$.
 20. **if** *accept* **then**
 21. $\mathcal{A}_2[c] \leftarrow \mathcal{A}_1[c]$; Increment *acceptance_counts*[c].
 22. **else if** $p \geq \alpha$ **then** Increment *rejection_p_count*[c].
 23. **else** Increment *rejection_es_count*[c].
 24. **return** $\mathcal{A}_2$
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Algorithm 2: HT Replacement Rule

1. **Input:** $r^{\text{new}}, r^{\text{old}} \in \mathbb{R}^M$, significance α , effect size floor $\delta_{\min}$.
 2. Calculate means ($\bar{r}^{\text{new}}, \bar{r}^{\text{old}}$) and variances ($s_{\text{new}}^2, s_{\text{old}}^2$).
 3. *// CLES (Probability of Superiority)*
 4. $\hat{\delta} \leftarrow \frac{1}{M^2} \sum_i \sum_j [\mathbb{I}(r_i^{\text{new}} < r_j^{\text{old}}) + 0.5\mathbb{I}(r_i^{\text{new}} = r_j^{\text{old}})]$.
 5. *// Exact Z-test Standard Error of Difference*
 6. $SE \leftarrow \sqrt{\frac{s_{\text{new}}^2}{M} + \frac{s_{\text{old}}^2}{M}} + \epsilon$.
 7. $z \leftarrow \frac{|\bar{r}^{\text{new}} - \bar{r}^{\text{old}}|}{SE}$.
 8. $\hat{p} \leftarrow 2(1 - \Phi(z))$ *// Exact Gaussian CDF*
 9. **Result:** accept $\leftarrow (\bar{r}^{\text{new}} < \bar{r}^{\text{old}}) \wedge (\hat{p} < \alpha) \wedge (\hat{\delta} > \delta_{\min})$.
 10. **return** (accept, $\hat{p}$, $\hat{\delta}$)
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5. Experimental Setup

5.1. Simulated Environment

To model the stochastic nature of model feedback defined in Eq. 1, we implement a synthetic evaluator pool ($N_V = 10$).

- **Perceptual Feature Space:** We extract five features $\psi(x) \in [0, 1]^5$ representing Brightness, Contrast, Entropy, Edge Density, and Symmetry. These serve as proxies for the high-dimensional aesthetic criteria used by real-world VLMs.
- **Model Jury:** Each model v in the pool is assigned a fixed preference vector w_v . To ensure heterogeneous but comparable evaluations, weights are sampled as $w_v = |\mathcal{N}(0, I)|$ and subsequently L_2 -normalized.
- **Noise Regime:** We set $\sigma = 0.05$ and $M = 10$, creating a high-noise environment where the standard error of the difference ($SE = \sqrt{\frac{\sigma_{\text{new}}^2}{M} + \frac{\sigma_{\text{old}}^2}{M}}$) necessitates robust statistical intervention.
- **Deterministic Phenotype Map:** To ensure each candidate has a well-defined true quality, the phenotype map $\psi(\cdot)$ is a fixed deterministic projection of the genotype ($\psi(g) = \sigma(g \cdot P)$ where P is a fixed random projection matrix and σ is the sigmoid function). Evaluation noise $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ is therefore the sole stochastic source, matching Eq. 1 exactly and enabling an analytical ground-truth fitness $f_{\text{GT}}(x) = -\mathbb{E}[r(x)]$ with no sampling required.

5.2. Evaluation Protocol

We evaluate HT-MAP-Elites across $T = 300$ iterations using $N = 15$ independent random seeds. The experiment utilises a 5×5 behavioural grid. Beyond the baseline noise level ($(\sigma) = 0.05$), we sweep $\sigma \in 0.05, 0.10, 0.20, 0.30$ to characterise behaviour across noise regimes. For each finalised archive we compute an analytical ground-truth QD score using the noise-free true fitness of each occupied elite, and define the inflation gap as $\Delta = \text{QD}_{\text{apparent}} - \text{QD}_{\text{GT}}$.

To address the ‘‘Zero-Progress’’ trap, we depart from a temporal exploration window and instead utilize the concurrent *Leader-Follower* architecture described in Section 5. The Scout Archive (A_1) provides continuous exploration through naive mean-based replacement, while the Judge Archive (A_2) performs distributional refinement every $f_{\text{refine}} = 5$ iterations to maintain a robust and verified repertoire.

5.3. Hyperparameters

Table 1: Updated Hyperparameter Settings

Symbol	Meaning	Value
<i>Environment & Evaluation</i>		
N_V	Total model pool size	10
M	models sampled per evaluation (Rater count)	10
σ	model rating noise standard deviation	0.05 (baseline)
w_v	model preference vector ($ \mathcal{N}(0, I) , L_2$ -norm)	Fixed, Seed 42
$\psi(x)$	Perceptual feature vector ($\in [0, 1]^5$)	—
<i>Hypothesis-Testing Rule (HT-MAP-Elites)</i>		
α	Significance threshold for Z -test	0.10
$\delta_{\min}$	CLES floor (Practical significance)	0.55
p	Exact two-tailed Gaussian p -value	$2(1 - \Phi(z))$
$\hat{\delta}$	Common Language Effect Size (CLES)	Eq. 5
<i>Quality-Diversity Search</i>		
T	Total iterations (Concurrent run)	300
f_{refine}	Refine Frequency (Sync interval for A_2)	5
B	Batch size	8
N_{seeds}	Independent random seeds for paired testing	15

6. Results

We conducted a paired comparative analysis between the naive Scout archive (A_1) and the robust HT-MAP-Elites Judge archive (A_2) across $N = 15$ independent random seeds, each running for $T = 300$ iterations.

Table 2: Final Hyperparameter Settings for HT-MAP-Elites Evaluation

Category	Hyperparameter	Value
HT rule	Significance Threshold (α)	0.10
HT rule	CLES Floor ($\delta_{\min}$)	0.55
Environment	Evaluation Noise (σ)	0.05
Environment	Rater Count (M)	10
QD Search	Total Iterations (T)	300
QD Search	Refine Frequency (f_{refine})	25
QD Search	Independent Seeds (N_{seeds})	15

6.1. Score Inflation in Naive MAP-Elites

We first establish that naive mean-based replacement accumulates stochastic artifacts. For each finalised archive we compute the inflation gap $\Delta = QD_{\text{apparent}} - QD_{\text{GT}}$, where QD_{GT} uses the analytical true fitness per elite. The naive archive A_1 exhibits a strictly positive inflation gap that grows monotonically with noise: 0.43, 1.11, 2.90, and 4.62 QD units at $\sigma \in \{0.05, 0.10, 0.20, 0.30\}$ respectively. This confirms that lucky noise draws are erroneously locked into the archive over the course of evolutionary search, inflating its apparent quality above its true quality.

Table 3: Apparent vs analytical ground-truth QD across noise levels ($N = 15$ seeds, $T = 300$, $B = 8$, $f_{\text{refine}} = 5$). Inflation = Apparent – Ground-Truth. A2’s inflation is significantly lower than A1’s at every σ (paired t-test, all $p < 0.05$).

σ	App. A1	App. A2	GT A1	GT A2	Infl. A1	Infl. A2
0.05	36.38	35.38	35.95	35.03	0.43	0.35
0.10	35.86	34.79	34.75	33.87	1.11	0.92
0.20	35.36	33.83	32.45	31.37	2.90	2.46
0.30	35.52	33.68	30.90	29.67	4.62	4.01

6.2. HT Gate reduces Inflation

The HT archive A2 exhibits a significantly smaller inflation gap than A1 at every noise level (paired t-test: $p = 0.034$, 1.6×10^{-4} , 1.7×10^{-4} , 1.6×10^{-5} for increasing σ ; Table 3). The gate therefore produces a repertoire whose reported scores are a more reliable estimate of true quality — valuable precisely when ground-truth re-evaluation is unaffordable, as with real VLM evaluators.

6.3. The Conservatism Trade-off

This reliability comes at a cost. Under single-evaluation gating, A2’s ground-truth QD remains below A1’s at every noise level (Figure 1a), because the gate rejects 93–96% of candidates (Figure 1c), including genuine improvements it cannot distinguish from noise

with only $M = 10$ ratings. The dominant failure mode is the p-value test (Figure 1c), indicating insufficient statistical power rather than effect-size filtering. Figure 1d confirms the ground-truth gap is statistically significant at every σ .

6.4. Evaluation Budget Controls the Trade-off

The conservatism is not intrinsic to the HT rule. When A2 pulls a candidate from A1, the candidate’s cached scores carry a selection bias (the winner’s curse): A1 admitted it because those scores were favourable. Re-evaluating with independent noise draws removes this bias. Increasing the number of independent re-evaluations K at the gate raises the gate’s statistical power, allowing it to admit genuine improvements while keeping inflation low. The exploration–validation trade-off is thus governed by the evaluation budget $K \times M$ spent at the gate, rather than by the gate rule itself.

7. Discussion

The experimental data validates the core hypothesis of HT-MAP-Elites: that decoupling exploration from verification stabilizes the repertoire against rater noise.

7.1. Interpreting the QD Score Gap

We resolve the question of true archive quality via analytical ground-truth evaluation. The naive archive’s positive, noise-scaling inflation gap (Section 6.1) confirms that its apparent QD advantage is partly an artifact of lucky noise draws being locked into the archive. The HT gate significantly reduces this inflation (Section 6.2). Under single-evaluation gating, however, the HT archive’s ground-truth QD remains below the naive baseline’s, because the gate — lacking sufficient statistical power at $M = 10$ — rejects the majority of candidates including genuine improvements (Section 6.3). We show this conservatism is a budget effect, not an intrinsic limitation: independent re-evaluation at the gate removes the selection bias and additional re-evaluations recover true quality (Section 6.4).

7.2. The Role of Refine Frequency in Acceptance Rates

At the refine frequency used in this study ($f_{\text{refine}} = 5$), the Judge sees relatively immature candidates and the gate is active, rejecting 93–96% of attempts. This exposes the gate’s true filtering behaviour. At higher f_{refine} (e.g., 25), the Scout pre-filters candidates for many iterations before they reach the gate, which can produce near-100 % acceptance — not because the gate is ineffective, but because pre-filtering has already removed the weakest candidates.

7.3. Preservation of Diversity

Crucially, the HT-MAP-Elites HT replacement rule does not come at the cost of diversity. Both archives achieve full behavioural coverage (100%) across all seeds and noise levels. The HT gate’s conservatism therefore affects which elite occupies each niche, not whether niches are filled: it does not impede behavioural diversity.

8. Limitations and Future Work

Firstly, the experimental environment relies on a synthetic model jury with controlled Gaussian noise and fixed perceptual features, which, while enabling rigorous analysis, does not capture the full complexity of large scale generative models such as VLMs which may have prompt sensitivity, model versioning, and non-stationary preference drift. Deployment on large scale generative models is therefore a logical next step for the evaluation of HT-MAP-Elites.

Second, the statistical testing framework assumes that evaluator scores are approximately normally distributed and mutually independent across raters. In practice, generative models trained on overlapping data may exhibit correlated scoring behaviour, and score distributions may be skewed or heavy-tailed, which could affect the validity of the Z -test and CLES estimates. To counter these, various non-parametric hypothesis tests could be used which do not require normality assumptions.

Third, under single-evaluation gating the HT archive trades true quality for score reliability. Recovering both simultaneously requires additional evaluation budget — independent re-evaluations K at the gate — which we characterise (Section 6.4) but do not exhaustively optimise. The optimal K is expected to be problem- and noise-dependent when scaling to new domains.

Last, tightening the gate thresholds (e.g., raising $\delta_{\min}$ from 0.55 to 0.65, or reducing α from 0.10 to 0.05) may similarly induce rejections and would clarify the sensitivity of the framework to its hyperparameters. Such an ablation remains an open direction, particularly as these values are expected to be problem-dependent when scaling to new domains or noisier evaluation regimes.

9. Potential Applications

9.1. Accelerated Drug Discovery and Molecular Design

In chemistry, molecular properties like binding affinity and toxicity are often predicted using noisy simulations or ensemble ML models. HT-MAP-Elites can stabilize the search for diverse chemical scaffolds by treating uncertain predictions as distributions, ensuring a candidate molecule only replaces an incumbent lead when its superior efficacy is statistically significant. This would also prevent the loss of high potential "elite" compounds, allowing researchers to maintain a high quality repertoire of diverse molecular candidates for further clinical validation.

9.2. Personalized Content Recommendation Systems

In recommendation engines, user engagement serves as a noisy, multi-rater evaluator. By modeling feedback from diverse demographics as a rating distribution (r), HT-MAP-Elites’s replacement rule prevents “lucky” but low-quality assets from erroneously replacing stable elites. This statistically-grounded selection preserves a diverse repertoire of high-performing content, making the system resilient to engagement spikes and noise-driven churn. Consequently, the archive maintains genuine personalization that reflects consistent user preferences rather than ephemeral, stochastic fluctuations.

9.3. Generative model alignment

In creative AI pipelines, such as those powering Midjourney or DALL-E, models are increasingly aligned with human preference using Vision-Language Models (VLMs) as proxies. However, scoring is extremely subjective; a single image can receive widely varying scores across different VLM versions. HT-MAP-Elites’s judge archive ensures that a high-quality “elite” image in a specific niche is not displaced by a noisy, high scoring candidate that lacks genuine aesthetic superiority. The alignment process is more stabilised, ensuring that the generative trajectory stays focused on statistically verified improvements.

10. References

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